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A systematic approach for restoring sustainability in the hilly regions of northeast India: the Zabö conceptual framework

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Abstract Social, environmental, and economic problems are the major concerns for the agro-ecological communities. To address these environmental and economic threats, it is essential to understand how individuals think and act towards the existing practices working for these agricultural heritage systems, in the changing times, to preserve them. There are various indigenous agriculture practices for the varied particular regions, which have been practiced for centuries in their respective ecosystems. These indigenous practices include the knowledge of the environment of the specific area in a deep sense, and locals utilize them to coexist with nature by preserving the environment and fulfilling human needs. Therefore, there is a need to recognize the agricultural heritage to bring sustainability and economic stability to the agroecological regions. One such region is Zabö, a potential AHS (Agricultural Heritage System) site in northeast India, also known as Rüza. Thus, this paper seeks to address the multidimensional factors of sustainability and vulnerability factors in the context of climate change impacts via an area study. Specifically, the study focuses on aligning the GIAHS framework with a long-term conceptual approach for the preservation and recognition of the Zabö AHS, as it holds the answer to sustainability in the long-term traditional, ecological, and socio-cultural knowledge and values.

Keywords Agriculture heritage sites · Conservation · GIAHS · Sustainability · Zabö

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Introduction

The human civilisation needs new models of agriculture in the near future, which are more biodiverse, local, resilient, sustainable, and socially just (FAO) in nature. Sustainable development is one of the dual focuses of the Millennium Development Goals (MDGs); the second focus outlines the steps required to achieve global sustainability. The new sustainable agriculture models are aligned with the Sustainable Development Goals and the MDGs targets, highlighting the need to identify Globally Important Agricultural Heritage Systems (GIAHS) (Zhao et al., 2021). To achieve these goals, the agricultural system must be reoriented to adapt to a rapidly changing physical and economic environment.

Agriculture systems, which are unique identities of the physical environment in certain regions, are designated as Globally Important Agriculture Heritage Systems (GIAHS) by the Food and Agriculture

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Organization (FAO). Africa has 4 GIAHS, 1 in Kenya, 2 in Tanzania, and one in São Tomé and Príncipe. Asia and the Pacific have 65 across 10 countries, 1 in Bangladesh, 25 in China, 3 in India, 6 in Iran, 17 in Japan, one in the Philippines, 9 in the Republic of Korea, 1 in Sri Lanka, 1 in Thailand, and 1 in Indonesia. Europe and the Central Asian region have 15 GIAHS across 6 countries: 1 in Andorra, 2 in Austria, 3 in Italy, 6 in Spain, 1 in Tajikistan, and 2 in Portugal. Latin America and the Caribbean have 11 GIAHS across 5 countries: 2 in Brazil, 3 in Chile, 1 in Peru, 3 in Mexico, and 2 in Ecuador. East and North Africa have 9 GIAHS across 5 countries: 1 in Algeria, 3 in Morocco, 3 in Tunisia, 1 in the United Arab Emirates, and 1 in Egypt, in addition to several proposed sites across the world. The designation helps in preserving the disappearing sustainable practices, which hold the potential benefits for the local economy in the long run (Jiao et al., 2022). India, despite having diverse physiographic regions and environments, has only three designated GIAHS sites and no proposed sites for the future. Therefore, the study adopts a forward-looking approach to examine why conservation must be aligned with the goals of the GIAHS framework.

Despite the growing global recognition and policy momentum of GIAHS, Indian agri-system remains conspicuously under represented, especially the northeast Indian indigenous system. Zabö, despite having been sustained for close to a century in Kikruma (Das Barasha, 2023), remains one of the few documented integrated water-harvesting systems in the eastern Himalayan foothills, has received no attention in the domestic and international literature, and no study has evaluated it against FAO selection criteria or situated it within a policy actionable conservation framework. This neglect likely reflects the system's small operational scale relative to more prominent GIAHS candidates, only 26 of Kikruma's 915 hectares of agricultural land are currently under zabö cultivation and its absence to date from any state or national conservation inventory, both of which have kept it outside the scope of existing GIAHS assessments in India despite its ecological and institutional distinctiveness. Therefore, this paper addresses that gap. Specifically, it examines how Zabö can be assessed as a potential GIAHS through a spatially grounded, dynamic conservation framework that links landscape changes, indigenous water governance, livelihood vulnerability and multi-level policy support. With this objective, this study aims to (i) characterise the geographical, ecological, and social-cultural dimension of the Zabö system in Kikruma, (ii) assess its alignment with FAO GIAHS selection criteria, (iii) identify the climatic and land-use vulnerabilities threatening its continuity, and (iv) propose a multi-level (international to local) policy framework for recognition, dissemination and adaptive management for dynamic conservation.

Need for dynamic conservation amid globalization and how?

Traditional agriculture systems at present serve the plate for approximately two billion people globally (FAO, 2017b; Hak-Yeol, 2016).The system sustains biodiversity, culture, livelihoods, and practical knowledge, thereby enabling the system to continue providing for the needs of present and future generations sustainably. These complex adaptive systems emerge when natural ecological processes and human interactions are combined to create complex adaptive systems, which produce a mutually supportive agro-ecological system (Fuller et al., 2015), and are commonly referred to as AHS. Moreover, these represent complex and unique adaptive systems and vary across the regions according to the physical geography and human approach.

In recent times, these ecological systems have consequently been affected due to the growing demand and climate change, alongside the shift in farming approaches. Traditional agroecosystems have been consequently adversely affected as agricultural paradigms have increasingly prioritized agricultural intensification (Arnés García et al., 2020). Moreover, agriculture cannot be considered sustainable unless it is lucrative and capable of providing farmers with a good standard of living and income, preventing irreplaceable loss to socioeconomic capital (Rao et al., 2019; Mishra et al., 2022). Agriculture serves as the only means of livelihood, which contributes economically to the many Indigenous communities at the primary level. GIAHS is much more than a traditional Heritage site, which is a protected region and landscape. Evolved over many generations to accommodate specific local environmental circumstances, many agricultural heritage systems are innately resistant to climate change to a great extent. These systems

can aid in the mitigation of climate change impacts and thus act as role models for agricultural practices that are climate resilient in nature (Shamas et al., 2019). Therefore, the main objective of the GIAHS Programme is to identify and provide systematic support for the conservation and adaptive management of agricultural heritage systems. The GIAHS Program promotes intervention strategies at global, national, and local levels for dynamic conservation, sustainable management, and increased viability at all levels.

Global level—Global recognition of AHS is achieved through the recognition of unique agricultural heritage by the government at all levels. Within the framework of the United Nations Collaborative Program on Reducing Emissions from Deforestation and Forest Degradation (UN-REDD), the conservation of AHS forestry can be effectively addressed. This brings awareness towards the understanding of culture and climate-led threats, alongside compiling and sharing best practices and lessons gained from AHS around the globe.

National level—The countries China, Japan, and Korea have given recognition to the agricultural heritage systems, and thus their project activities guarantee sectoral and inter-sectoral plans and policies that embrace the GIAHS concept at the mainstream level. Action plans like Nationally Appropriate Mitigation Action (NAMAs) can actively be part of the plan for Agriculture, Forestry and Other Land Use (AFOLU), as it aims to reduce carbon emissions in developing nations. This could impact the development of NAMAs, which are linked to measurable emission reductions, and could therefore induce political and economic policy changes that better support traditional agriculture (Altieri et al., 2017). Thus, in regard to capacity building at the ground level and ensuring the food security along with poverty alleviation, interventions from national and local government institutions are needed.

Local level—The need for action at the local level acknowledges community-level adaptive management and conservation, along with the increase in economic support. Organizations (North East Rural Livelihood Project (NERLP) and Self-help groups) and representative committees need to establish an understanding for small farmers to ensure traditional agriculture and food security(Altieri et al., 2017). Globalization has significantly altered the trajectories of the landscape (Min et al., 2019). Therefore, to halt the rapid degradation of AHS, their dynamic nature must be recognized first. Their resilience depends on their ability to adapt to new challenges without losing their biological and cultural wealth and their productive capacity (Bridgewater et al., 2019). Upholding without making steps towards conservation with an integrated approach would undoubtedly cause them to deteriorate and plunge the populations into poverty (Dela-Cruz et al., 2009).

The AHS approach is centred around human management-based knowledge systems, including their resilience, sustainability, and integrity, while maintaining the cultural, economic, and socio-organizational elements that support the conservation and adaptation processes in GIAHS (Barnes, 2010; Lun et al., 2021). Prior empirical evidence from north-eastern India shows that nature based participatory approaches can strengthen biodiversity knowledge and human nature connectedness, when the local communities are engaged as co-learners rather than passive recipients of conservation awareness (Ghosal et al., 2023). This supports the need to treat knowledge transmission and community participation as central components of dynamic conservation in agriculture heritage systems. According to the development plan proposed by FAO, the objective to achieve the goals is to give enough attention at the international and national levels, along with institutional support for its preservation (Jacob, 2015). To elevate the benefits, labelling is a helpful way to promote the indigenous systems, i.e., Eco tourism, Environmental service, and Eco labelling, and to promote regulatory policies and incentive environments to support the conservation, evolutionary adaptation, and viability of AHS (FAO, 2017a). Therefore, for the assessment of the existing policies in the region, incentive-based new regulations are required to improve AHS economically and ecologically. A similar approach has been taken by other nations as well. In 2012, concrete steps were taken by MOA (the Ministry of Agriculture and Rural Affairs of the People's Republic of China) in this direction by designating NIAHS in China (Jiao et al., 2017). To bring change in the GIAHS, countries like China, Japan, and Korea also established national-level schemes. In the year 2016, MAFF (Ministry of Agriculture, Forestry and Fisheries) established the Japanese Nationally Important Agricultural Heritage Systems (J-NIAHS), and it was effectively successful in the conservation of

designated sites, along with contributing to achieving global goals of sustainable development (Akira et al., 2021). Not only the formation of the organization but also the adaptation to Agri-tourism is a great fit for the conservation of agricultural heritage (Jiao et al., 2016; Kajihara et al., 2018).

In China, Agro-tourism has played an important role in generating sustainable income(Yang et al., 2018).In Korea, the introduction of theme parks and trails for the JBAS (Jeju Batdam Agricultural System) has been constructed for environmental improvement in the short term (Province, 2013). Along with it, these nations provide compensations and participation schemes for maintaining sustainability and the local social structure, where China provides GIAHS compensation (Liu et al., 2018), Korea has incentives for the well-managed AHS sites (Province, 2013), and Japan does not have a separate fund for GIAHS sites, however it does have committees at the municipal level to represent and for lookout (Akira et al., 2021). Moreover, if SLF (Sustainable Livelihood Framework) is applied to the AHS, it can comprehensively integrate the SD under the UNDP framework (Elizondo et al., 2017), identifying suitable strategies to enhance community engagement and improve community livelihood sustainability (Su et al., 2018). Hence, these regions have effectively promoted and protected AHS, and their approaches can therefore be adopted for the conservation of AHS sites such as Zabö in India.

Existing agricultural heritage system in India

The FAO sees the potential for the global community to fulfil the Millennium Development Goals (MDGs). Biodiversity conservation is given top priority by several agricultural heritage systems. These systems frequently entail the raising, protecting, and preserving of a variety of crops and livestock varieties. In addition to preserving genetic diversity, the system promotes the health of the ecosystem by fostering overall growth. Thus, the sites that have been chosen are anticipated to have a high level of agricultural productivity and biological biodiversity. However, both criteria are currently experiencing unique management challenges (Koohafk et al., 2010), although GIAHS recognition brought attention to these sites. An overview (Table 1) of the existing Agriculture Heritage System in India, along with its importance and changes in the socio-environmental status, has been provided. Sites in India that highlight the rich diversity and potential of indigenous practices as GIAHS include the saffron heritage of Kashmir, the Kuttanad farming system in Kerala, and traditional agriculture practices in Koraput, Odisha (Fig. 1). Understanding these systems can result in a constructive change in similar regions.

Saffron Cultivation: With the GIAHS designation, the decline in saffron cultivation has been reversed, ensuring its sustainability and the financial security of the practicing farmers. Kuttanad Wetland: Preservation of this distinctive system, conservation of biodiversity, and improvement of livelihoods for communities that depend on it have been aided by recognition of degradation. Koraput Traditional Agriculture: By promoting agritourism and giving local farmers more income opportunities, the GIAHS tag started to get the help that ensures the economic sustainability of the region.

Vulnerability to climate change

Unfavourable climatic patterns pose a significant threat to marginalized sections of society, disrupting their traditional way of living (Leary, 2008; UNDP, 2007; Wolf et al., 2010), as these communities are largely dependent on informal livelihoods. Moreover, developed and developing countries will see the addition of a new poor population between the present and the year 2100, led by climate change-induced poverty, compromising sustainable development (Olsson et al., 2014). Therefore, recognition and acknowledgement are the first essential steps towards sustainability. Since AHS serves as archives for important resources for coping with climate change, such as traditional knowledge, genetic resources, and natural resource management techniques (Koohafk et al., 2010).

Assessing the related traditional knowledge is crucial for sustainable forest management as well as for its multipurpose role (Santoro et al., 2020). Moreover, climate change-based impact on the AHS determines the distribution of output over geographic regions and different latitudes, and the composition and geographical allocation of crops and types of livestock (Tubiello et al., 2008). With the increasing frequency of extreme events worldwide such as heatwaves and droughts in the

Table 1 Overview of existing and potential GIAHS in India

Site location	Saffron heritage (pampore) Kashmir	Koraput region, Odisha	Below sea level farming, Kuttanad, Kerala	Zabö farming, Nagaland ("Exploring Sustainability through Zabö: A Conceptual Framework" section)
Designation year	2011	2012	2013	Not yet designated
Core agricultural system	Saffron cultivation (crocus sativus) & agro-forestry	Multi-cropping tribal agriculture & traditional rice landraces system	Integrated paddy-fish-duck rotational farming in low lying polders	Integrated forest-pond- livestock terrace paddy agriculture
Physical & hydrological resources	Elevated karewa terrace plateau with well drained fertile soil	Fertile soil with abundant groundwater & biodiversity in the undulating highlands & valleys	Low-lying wetlands (0.6–2.2 m below sea level), garden lands, and backwaters	Fertile lands, steep hill slopes (> 20%), abundant rainwater, gravity-fed catchments with highland forest
Community & cultural practices	Traditional corm selection, manual soil aeration, and field management	In-situ seed conservation, traditional, tribal land rotation, and participatory biodiversity protection	Bund making, dewatering management& padasekhara committee coordination	Pond de-silting channel maintenance, clan-based water sharing, and integrated nutrition recycling
Area of GIAHS	32sq Km	500sq Km	Ca 900sq Km	65sq Km (26 ha under active cultivation in kikruma)
The Population Working for This System	17000 Farm Families	> 70% Tribal Population	No Information Available	1664 Families
Climatic classification	Temperate/sub mediterranean high altitude tropical conditions	Tropical monster with high seasonal conditions	Humid tropical coastal wetlands	Sub- temperate/moderate humid climatic conditions
Ethnic groups/indigenous people	Kashmiri agrarian communities	52 tribal groups	Traditional agrarian & fishing communities (kayal/farmers)	Chakhesang naga tribe
Primary income sources	Cash crop agricultural (saffron), horticulture (nuts & fruits)	Agriculture, non-timber forest products (NTFPs)	Paddy farming, fisheries, duck rearing, ecotourism	Integrated terraced agriculture, livestock husbandry, forestry, and gi-tagged rice
Observed Impact After Recognition	Stabilization of market prices, enhanced branding, increased & consistent supply, gi certification support	Created the niche market for the genetic crop seeds	Padasekhara (field) committees for maintenance & institutional strengthening	Not yet designated (baseline assessment observes high potential)
Socio-eco benefits	Preserved corm genetic diversity & strengthened economic state & incentivization	Safeguarded more than 340 traditional rice landraces, ppp collaborations	Regulated estuarine floodwaters, sustained unique below, sea level food production & ecotourism	Anticipated water security, biodiversity conservation, and diversified income upon recognition
Future scope & research direction	Ecological niche modelling for climate resilient saffron expansion in suitable regions	Introduction of courses at the state university on indigenous agrobiodiversity & seed conservation	To reduce vulnerability to climate change & seawater intrusion & tidal flooding	Expansion of policy recognition, gi market integration, monitoring & scalability to other eastern himalayan catchment systems (as per this study)

Table 1 (continued)

Site location	Saffron heritage (pampore) Kashmir	Koraput region, Odisha	Below sea level farming, Kuttanad, Kerala	Zabö farming, Nagaland ("Exploring Sustainability through Zabö: A Conceptual Framework" section)
Alignment with FAO GIAHS criteria	Food security & landscape distinctiveness (1,2,4,5)	Biodiversity & knowledge systems (1,2,3,4)	Food security & landscape distinctiveness (1,2,5)	Fully aligned as per study assessment (1,2,3,4,5). food and livelihood security (§3.2–3.3); biodiversity (§3.4, gi tagged crops); traditional knowledge (§3.5 water impounding); social organization (§3.3 chakhesang water sharing institutions); landscapes (§3.1 terraced hill topography)
Source	FAO GIAHS; local agri-survey records	FAO GIAHS, Raju et al., 2020; CGIAR, 2025	FAO GIAHS, Sumanapala & Wolf, 2026	Singh et al., 2018; Yadav and Singh (2021); Das and Barasha, 2023

(*Source:* GIAHS | FAO of the United Nations, n.d., and Nagaland State Portal, n.d.)

FAO GIAHS pillar key: 1 = Food and livelihood security; 2 = Agro-biodiversity; 3 = Local and traditional knowledge systems; 4 = Culture, value systems, and social organization; 5 = Landscapes and seascapes

Mediterranean region and intensified heavy precipitation and flooding in temperate zones (Lazoglou et al., 2024); Cusack et al., 2025). Europe has experienced recurrent heatwaves, particularly in southern and central regions (Bilgili et al., 2025). Flash floods driven by intensified rainfall have also become more frequent (Ludwig et al., 2023; Silva et al., 2025). In addition, Europe has witnessed megadroughts associated with precipitation deficits and elevated evapotranspiration (Tripathy et al., 2023; Bevacqua et al., 2024). In India, climate extremes are surging. Heatwaves in the north and central parts (Molina et al., 2026), drought across 65% of the entire country, and flash floods in post monsoon season (Shahi et al., 2023). The diurnal trend in the recent decade has seen a shift in the maximum and minimum temperatures in the NEI region (Paul et al., 2023). This will lead to evapotranspiration and affect the crop's health and yield.

On the other hand, climate change, in parallel with the existing structural inequalities and power balance, leads to multi-dimensional vulnerabilities, heightening the risk of climate extremes. This leads to substantially lower production and productivity of agriculture, along with the livestock mortality (Tubiello et al., 2008), reduction in food and water security (Romero et al., 2023). Recent hydrosocial research from eastern India shows that water scarcity and resource stress are often shaped not only by physical availability, but also by governance failures, uneven access, extractive pressures, and social inequalities (Ruj, 2025). This insight is relevant to Zabö because the system depends on collectively managed water harvesting. community-based sharing arrangements, and landscape-specific knowledge. Therefore, climate action through agroecological strategies is necessary to enhance the ecological resilience of farming systems; this alone is a necessary but not sufficient condition for achieving sustainability. The capacity of local communities to adapt in the face of external social, political, or environmental stresses must go hand in hand with ecological resilience. Likewise, Zabö AHS has already started to face the pressure of change in the system. Traditionally, AHS have developed effective mechanisms for coping with extreme climatic conditions. However, climate change is expected to place increasing pressure on these systems in the future.

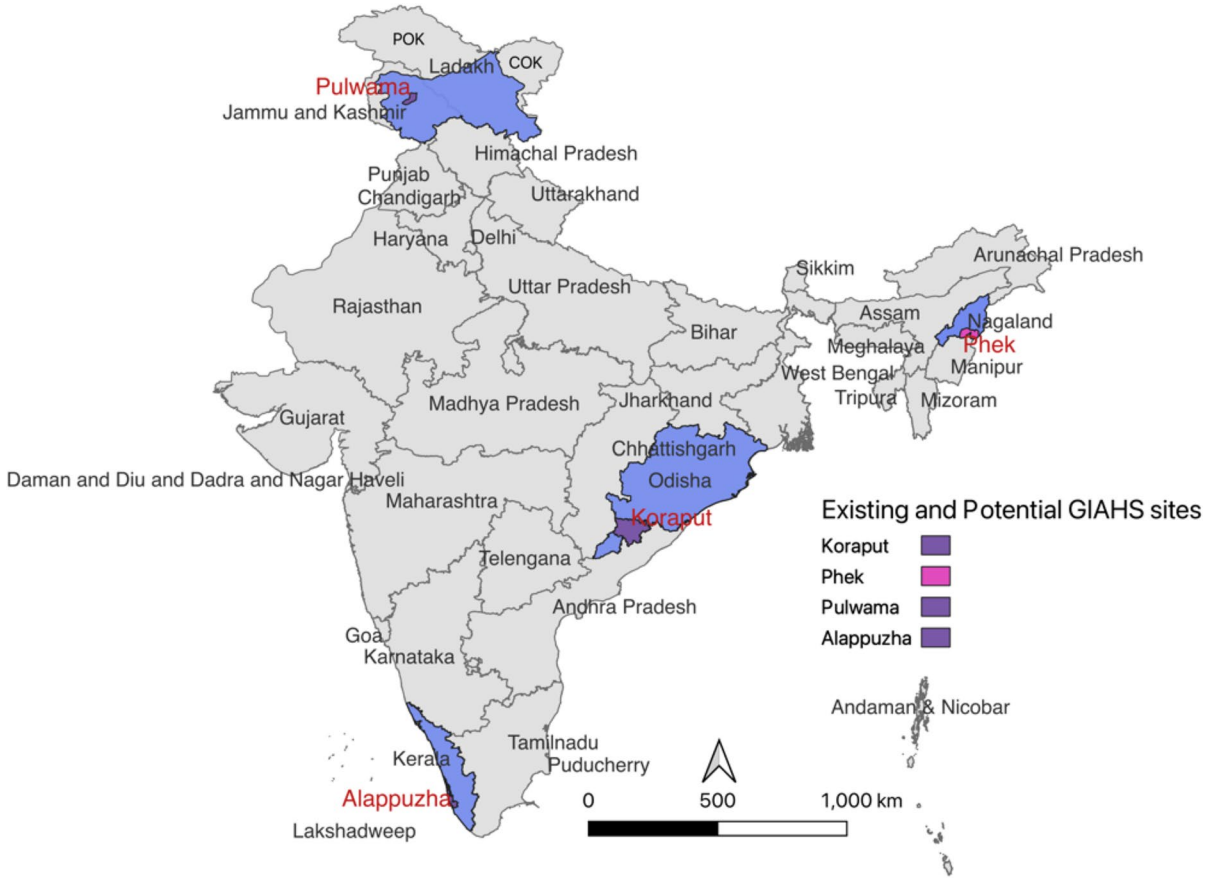


Fig. 1 GIAHS sites in India (existing and potential sites)

Materials and methods

In the study, at first initial area analysis was conducted, followed by the development of a conceptual framework into three sections. For the area study, LULC data access has been provided by the BHUVAN, ISRO. Software for mapping and visualization for the spatial analysis has been done using QGIS, Google Earth Pro, Surfer, and Quick Grid. For the conceptual framework, a collection of secondary qualitative data research has been done through government reports.

Spatial data acquisition was carried out in three stages. First, LULC rasters for the Kikruma study area were downloaded from BHUVAN (ISRO) for the year 2005 and 2020, at 1:25000 scale and reclassified into built-up, cultivated, forest, water body and rangeland classes to derive the change- detection layers. Secondly, aerial and satellite base imagery was sourced from Google Earth Pro to delineate the boundary of the zabö catchment pond terrace system and lastly terrain and elevation data were processed in QGIS to generate slope and elevation profiles and pond and settlement distribution was interpolated and visualized using Surfer and Quick Grid to produce peaks and depressions surface (Fig. 2). All spatial layers were projected to WGS84/UTM Zone 46N prior to analysis.

Zabö integrated farming system, a potential GIAHS site

The Indian subcontinent has a huge land mass (60.05% under the Agri sector) under agricultural practices, along with immense variation in the geomorphological and climatic distribution. However, many agricultural practices are not well known on

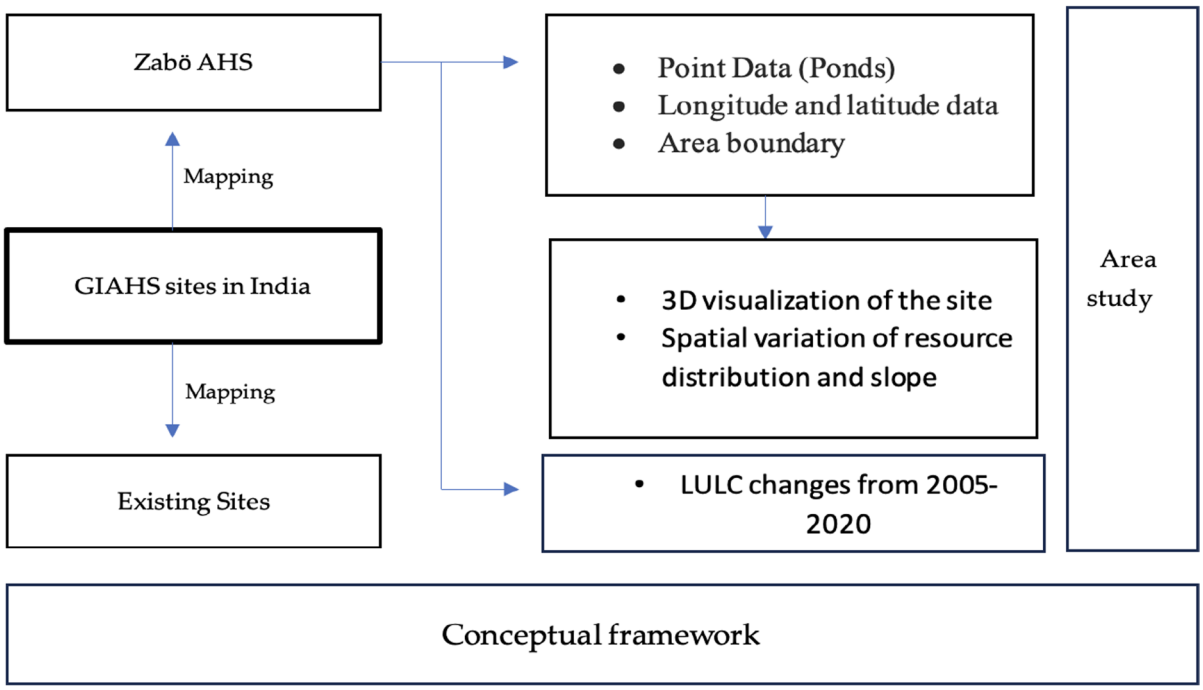


Fig. 2 Method and flow of the study

a larger scale, which possess the potential to be labelled as the future GIAHS. One of these kind AHS is Zabö, also known as Rüza, located in the eastern state of India, Nagaland. Both 'Zabö and local term 'Rüza, appear in the literature. This paper uses 'Zabö' throughout for consistency. The evidence base for assessment development in "Exploring sustainability through Zabö: a conceptual framework" section aims to draw on documented forest cover and biodiversity conservation ("Location, geography, and climate" and "Biodiversity" section), the terrace water harvesting sequence ("Farming system and its functioning" section), cultivated sticky- rice and GI-tagged crop varieties ("Biodiversity" section), and locally reported indigenous knowledge practices ("Sustainability of the system"), synthesized from published field studies and government sources as described in the methodology ("Materials and methods" section).

Location, geography, and climate

Kikruma is the representative of the Zabö integrated farming system, which is located in the Phek district of the Nagaland state (Fig. 3). The state of Nagaland is predominantly characterized by hilly terrain having slopes of more than 20% (CGWB, 2011a).

Phek is a mountainous region with 70% of its land covered in evergreen forest (TRI, 2018), situated in the easternmost part of north India. Kikruma is located at an elevation of 1270 m above mean sea level (AMSL). The highest elevation is 2400 m AMSL, which is on Zainbu mountain. In terms of climate, summers are warm but not oppressively so; the average temperature hovers about 27 °C, rarely rising above 32 °C. The monsoon season lasts from the end of May until the end of September. During January and February, the coldest months of the year, when the temperature drops below 0 °C, the Assamese plains nearby experience milder winters. There is 1,527 mm of rain on average every year.

Farming system and its functioning

Practices a unique and sustainable practice, known as "Zabö," which means impounding of water (Singh et al., 2012; Moudgil, 2014; Das, 2014; Zabö Farming System, 2023). The whole ecosystem of sustainability is a combination of integrated farming of Agriculture, fishery, and animal husbandry. The Zabö

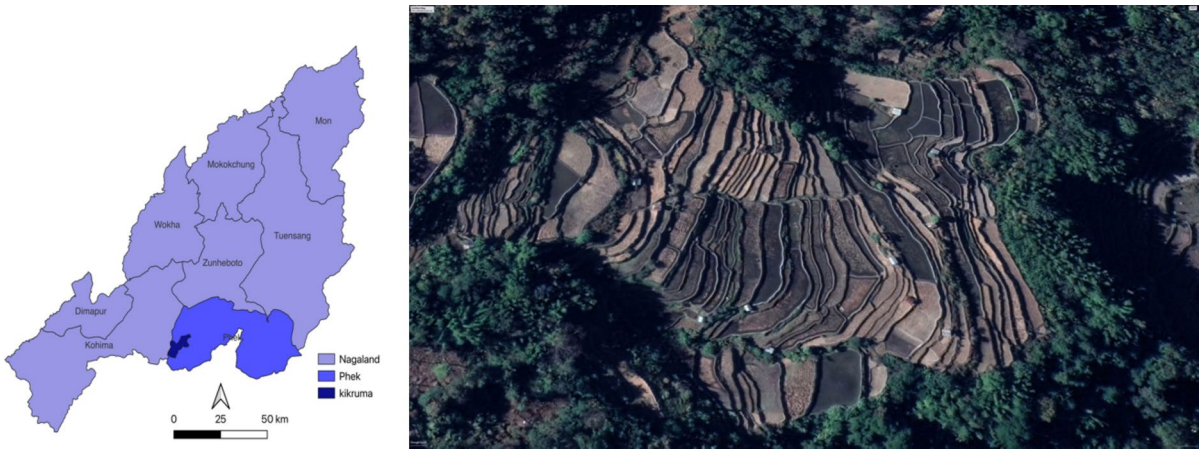


Fig. 3 Kikruma Location and Aerial view of a patch of Zabö farming in Kikruma region (*Source:* Google Earth image, updated on 1/2/2022 and accessed on 12/11/2023. Location coordinates are 25°35°06.86" N, 94●12′37.81"E)

farming system is made up of a complete continuum. The sequence of Zabö farming is as follows: on the highest elevation, permanent forest exists, along with the human settlement (Fig. 4a and b) Situated around 1600 m and above. Right after that, at a lower elevation, a catchment is constructed to directly collect the largest amount of water with a shallow depth of 1.5–2.5 m for the lean period (Fig. 4c). Moreover, the bottom surface of the pond is properly rammed to minimize the loss of water through seepage (FOCUS-IFAD, 2017). Catchments are constructed in such a way that the distribution of water is uniform. During summer, water from the catchment area is let into the rice fields located in the lower region, along with it carries waste and excrement of the animals on the way to increase the nutrient value of the field (Fig. 4e and f). After this, the water ultimately reaches the river basin or the lowest level of the system.

Water sharing and pond and channel repairs: Water harvested is shared across several families or clans through mutual agreements. Water travels via tunnels from one plot to the next until it reaches the last plot. They drain the extra water. Terrace culture sows from April to May, and transplants are done in June and July (Singh et al., 2012). Water is left in the fields for the duration of the plant's growth and then drained out right before harvest. By March or April, the ponds usually dry up, and repairs are made at this time. As the de-silted material has a good amount of organic matter and nutrients, it is delivered to the terrace fields below the pond. All of the households that have terrace fields in the irrigation command area take part in cleaning the siltation tank (Zabö Farming System, 2023).

Community and livelihood

The Naga community in Phek district lives at high elevations (Fig. 5), and fields exist at lower elevations. Agriculture is the main occupation, including Terrace Rice Cultivation (TRC) and shifting cultivation (Jhum Cultivation). Water scarcity is a challenge during the lean period in hilly areas, leading to the necessity of water management practices. Therefore, the community works closely with nature, sharing water resources through mutual agreements. Terrace culture is prevalent, with water shared through tunnels, and repairs made to ponds before the harvest season. Though Kikruma has 915 hectares of total agricultural land, only 26 hectares are farmed using the zabö technique. People who live at lower elevations and have access to enough water sources don't rely upon the catchment (Das Barasha, 2023).

Biodiversity

Being a hilly region, it is rich in flora and fauna. Thus, consisting of diverse vegetation, including evergreen forests. As per (Singh et al., 2018), in the Zabö farming system, 9 varieties of fruits, 14 varieties of vegetables, 6 varieties of sticky rice of different kinds, non-sticky rice of 6 different varieties, and 6 types of



Fig. 4 **a** Water flowing down from the high altitude to the lower altitude in a catchment, **b** Bamboo to control and check the soil erosion, **c** Evenly stamped land, **d** Water from the pond to the paddy field flows through the cattle shed, **e** Cattle shed, **f** Carrying the manure to the paddy field **g** & **h** Bunds to divert water **i** Extra water drained from the upper pond to the fishery and paddy field. (*Source:* Taken from various sources on the internet) *SOURCE:* (Yadav & Singh, 2021)

cereals and pulses, alongside 3 types of livestock and 3 types of water species exist. Moreover, there are three Geographical Indicator (GI) tagged vegetables from this region, i.e., Ich is Capsicum spp. (chilli), Cucumis Sativus (Cucumber), and *Sechium edule* (Chow Chow). Farming of a large variety of wild fish and prawns is also a micro part of the seasonal biodiversity (Das, 2018).

Sustainability of the system

There has been a decrease in the recharge and availability of water in the past decade (CGWB, 2011a; CGWD, 2022). During the monsoon season, recharge from rainfall was 1727, and during other seasons, recharge from rain was 784 cm. At the end, the groundwater availability for future irrigation was 218 mcm in 2011. However, this number dropped drastically within one decade to 539.16, 163.86, 590.36 Mcm in 2022, respectively. On the other hand, approximately one-third of the total area of Phek district is exposed to severe soil erosion. (Lohe et al., 2015). There are contributing factors to this phenomenon, including geographic conditions, slope, and high rainfall. Due to the small, elongated hills and other factors in the lean period, acute scarcity of drinking water occurs (CGWB, 2011b)as perennial sources of water are minimal along with abundant surface runoff (Das, 2014; Moudgil, 2014).

Therefore, the Naga community of the region works closely with nature to utilise natural resources to their fullest, such as the use of tree

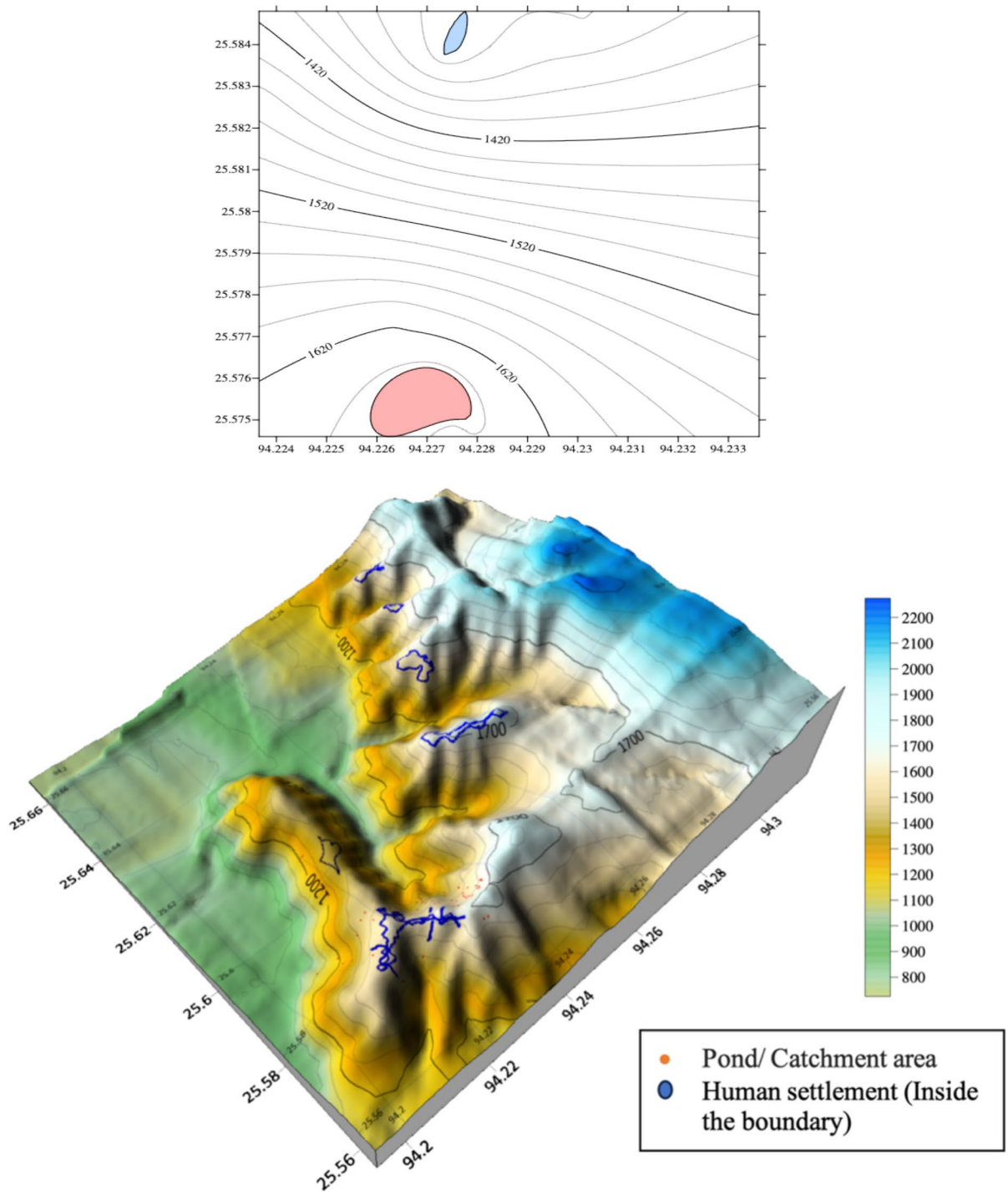


Fig. 5 Peaks and depressions in the region and Distribution of Zabö ponds and settlement in the Kikruma (Visualized using Surfer & Quick grid)

litter and common salt as fertilizer in minimal quantities, only when the paddy saplings are small (Das, 2023).

These local pressures on groundwater recharge and soil stability do not occur in isolation, documented shifts in maximum and minimum temperatures across the wider Northeast India region (Paul et al., 2023) compound the groundwater decline and soil erosion pressures already observed in Kikruma, reinforcing the case for treating climate variability and land use change as jointly determining the systems vulnerability rather than as separate factors. By contrast, socioeconomic pressures such as out migration of youth and intergenerational loss of Zabö specific knowledge are raised in this study as an observed concern requiring verification through future field-based research approach ("Conclusion" section), rather than as an evidenced finding, since no site-specific demographic or migration data were available within the secondary source scope of this study.

Within the past 5 years, there has been a massive change in the Kikruma region, as the majority of the changes took place in the built-up area and the rangeland. The built-up area expanded by replacing the permanent vegetation and increased with the changes in the extent of the cultivation land. Primary changes took place in the high-elevation (Fig. 6) along with the human settlement belt. Major changes can be noted in the built-up area and increasing multi-crop agriculture.

Exploring sustainability through Zabö: a conceptual framework

Theoretical positioning: Zabö as a socio-ecological and hydrosocial landscape

Agricultural heritage systems are increasingly theorised not merely as farming techniques but as bio-cultural landscapes, in which ecological processes, cultural practice, and institutional arrangements co-evolve over long time horizons (Bridgewater & Rotherham, 2019; Fuller et al., 2015). Within this framing, Zabö can be understood through several interlinked theoretical lenses relevant to geographical scholarship on sustainability and conservation.

First, the system exemplifies Indigenous ecological knowledge (IEK) as a form of situated, intergenerational environmental management, encoded in the sequencing of forest, catchment, cattle shed, and terrace field along the elevation gradient, and reproduced through community labour such as pond de-silting and channel repair. This knowledge also encompasses place specific understandings of hydrology, slope stability, seasonal cropping, and Chakhesang native understanding of water. Second, Zabö can be read through the lens of socio-ecological resilience, whereby the system's capacity to absorb disturbance and retain its core functions depends jointly on ecological buffering (impounded water, terracing, nutrient cycling via animal waste) and social buffering (mutual water-sharing

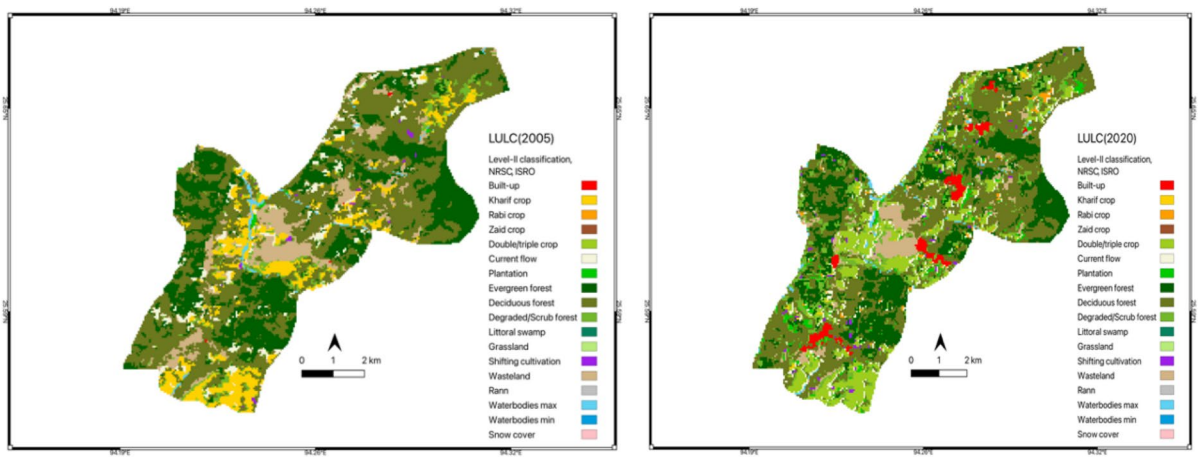


Fig. 6 Land use land cover change from 2005–2020, Kikruma (land use/land cover database on 1: 25000 scale, n.d.)

agreements, collective maintenance obligations among Chakhesang clans). Third, the system illustrates the principle of dynamic conservation central to the GIAHS approach, which rejects static preservation in favour of continued adaptations. Rather than being preserved as a static artefact, Zabö must continue to adapt in crop composition, water allocation, and built-up land use to remain viable under demographic, economic, and climatic change (Bridgewater & Rotherham, 2019). Fourth, and centrally to this paper's contribution, Zabö should be understood as a hydrosocial landscape rather than simply a water-harvesting technique. Water in this system is not a purely physical resource but a socially and institutionally governed one: its capture, distribution, and maintenance are structured by clan-based agreements, labour-sharing norms, and locally negotiated rights of access. These arrangements also raise concerns that top-down heritage recognition could bypass or weaken existing local institutions.

Recent hydrosocial scholarship from eastern India demonstrates that water governance is inseparable from questions of power, access, livelihood security, and community resilience, even in settings quite different from Zabö (Ruj, 2025). Framing Zabö in these terms shifts the analytical focus from "traditional technique" toward mountain landscape governance, the institutional and political arrangements through which highland communities manage scarce resources under conditions of ecological constraint and, increasingly, external development pressure. Finally, situating Zabö within a political ecology of conservation perspective draws attention to the uneven distribution of recognition, funding, and policy attention across India's agricultural heritage landscape. India's three designated sites are concentrated outside the Northeast, while a resilient, community-governed system such as Zabö remains absent from national policy instruments. A participatory approach is therefore important to ensure that recognition builds on rather than overrides, Chakhesang institutions and community based environmental knowledge. This theoretical positioning frames the remainder of the paper, it moves from description of the farming system toward an assessment of its heritage value and a proposed governance framework for its recognition and dynamic conservation.

Recognition, dissemination, and adaptation framework

The system's strengths, weaknesses, and opportunities frame this alignment. Strengths: Zabö is a low-cost, gravity-fed water management system requiring no external energy or chemical inputs; it integrates agriculture, animal husbandry, and fishery into a single nutrient-cycling continuum; it sustains high biodiversity (9 fruit varieties, 14 vegetables, 12 rice varieties, and three GI-tagged crops); and it is embedded in durable Chakhesang clan-based institutions for water sharing and collective maintenance. Weaknesses: The system is geographically constrained, only 26 of Kikruma's 915 hectares of agricultural land are currently under zabö cultivation, and the technique is viable only where topography and rainfall permit catchment construction. It also lacks a codified policy or institutional home at state or national level, leaving it dependent on informal, generational knowledge transmission that is vulnerable to outmigration and generational discontinuity. Documented declines in groundwater recharge and rising soil erosion further threaten the system's ecological base. Opportunities: GIAHS designation could unlock the compensation, agri-tourism, and eco-labelling mechanisms already demonstrated in China, Japan, and Korea; GI-tagged crops already produced in the region (chilli, cucumber, chow chow) offer a ready-made entry point for value-chain development; and existing but disconnected national and state schemes (NICRA, PKVY, MOVCDNER, TDF) could be formally extended to Zabö through the multi-level policy framework.

The recognition-dissemination-adaptation framework should therefore treat water management as a hydrosocial process, where sustainability depends on local institutions, equitable access, collective maintenance, and protection from externally driven land-use pressures (Ruj, 2025).

Further in this section, the exploration of the concept of sustainability has been done through the case study of Zabö, thus developing a framework for a holistic, long-term sustainable approach. This framework has been divided into three measures: Recognition, Dissemination, and Adaptation. Firstly, to recognize the site and its need for preservation, it should match all the parameters by FAO for GIAHS site selection. Therefore, looking at the primary objective, ZIFS (Zabö integrated farming system) shows

strong potential alignment with the selection criteria of GIAHS. ZIFS contributes to social welfare, conserves the unique biodiversity on the mountainous terrain (has a wide variety of rice and regional vegetables), passes on the unique integrated farming practices, holds community through pond making and water sharing practices, and lastly has unique mountain topography and ecological resources.

Secondly, to effectively coordinate these unique resources, an Analytic Hierarchy Process (AHP)-based decision-support engine is introduced within Stage 2 (Dissemination) to evaluate and prioritize protective policy instruments. AHP enables decision-makers to quantify subjective expert judgments and reconcile conflicting priorities across multi-level governance actors (ranging from local farmers and village panchayats to state ministries and international bodies). By evaluating pairwise comparisons and verifying mathematical consistency ($CR < 0.10$), this multi-criteria approach builds consensus across heterogeneous stakeholder groups operating under distinct institutional and environmental pressures. In this study, AHP is presented as an analytic structure for future prioritization rather than as a completed pairwise comparison exercise. To create a fully operational workflow, the framework establishes three proposed methodological engines linking its primary stages: (1) a spatial and risk-assessment engine (GIS-based LULC classification and an IPCC-style vulnerability index) supporting Stage 1; (2) the multi-criteria decision-support engine (AHP/Fuzzy-AHP pairwise comparison) supporting Stage 2; and (3) a participatory-impact engine (participatory GIS and pre/post comparative evaluation) supporting Stage 3. As with AHP, these represent the analytical structure this framework is designed to support rather than analyses completed within the present study; their execution is identified as a primary direction for future research.

At the international level, the policy recommendation aims to align the SDGs with key stakeholders, including the Global Environment Facility (GEF), which provides funding to NGOs and community-based initiatives. This alignment involves collaborating with SDGs and supporting global initiatives for sustainable agriculture and environmental conservation. Moving to the national level, the focus is on empowering Self-Help Groups (SHGs) for rural development, with key stakeholders like the Ministry of Environment, Forest and Climate Change (MoEFCC) and the International Fund for Agricultural Development (IFAD) acting as multi-stakeholder in various policies, alongside academic inclusion in National Education Policy and National Innovations on Climate Resilient Agriculture (NICRA) network project by the Indian Council of Agricultural Research (ICAR). Nevertheless, it does not include Zabö at present. These entities coordinate national policies related to environmental conservation and partner on sustainable agriculture projects. Such as to enhance the productivity of the fisheries sector and promote, the Fisheries and Aquaculture Infrastructure Development Fund (FIDF) by the Ministry of Fisheries, Animal Husbandry, and Dairying can help in setting heritage sensitive interventions through conservation compatibility through market support.

At the state level, key stakeholders such as State Biodiversity Boards (SBBs), the National Mission on Sustainable Agriculture (NMSA), and initiatives like Paramparagat Krishi Vikas Yojana (PKVY) and Mission Organic Value Chain Development in North Eastern Region (MOVCDNER) are responsible for implementing sustainable agriculture practices and biodiversity conservation. Alongside North East Slow Food and Agrobiodiversity Society, the Watershed Development Fund (WDF) holds multiple funding stakeholders. State-level initiatives primarily focus on implementing national policies and promoting sustainable agriculture within local communities.

Finally, at the local level, stakeholders like the Eco Mark Scheme, On Farm Water Management (OFWM), and Soil Health Card (SHC) play crucial roles in promoting environmentally friendly products, enhancing water efficiency in agriculture, and encouraging soil conservation and sustainable farming practices, Tribal Development Fund (TDF) by Ministry of Tribal Affairs (MoTA) all together holds the potential to holistically answer the regional and environmental stewardship for vulnerabilities through an inclusive approach. Local programs and schemes integrate with state-level initiatives to ensure sustainable agricultural practices and environmental conservation at the grassroots level. This hierarchical structure emphasizes alignment with SDGs across different levels of governance, with targeted support from international to local stakeholders. Neglected and Underutilized Species policy aim is covered under various policies like Rashtriya Krishi Vikas Yojana (RKVY) and

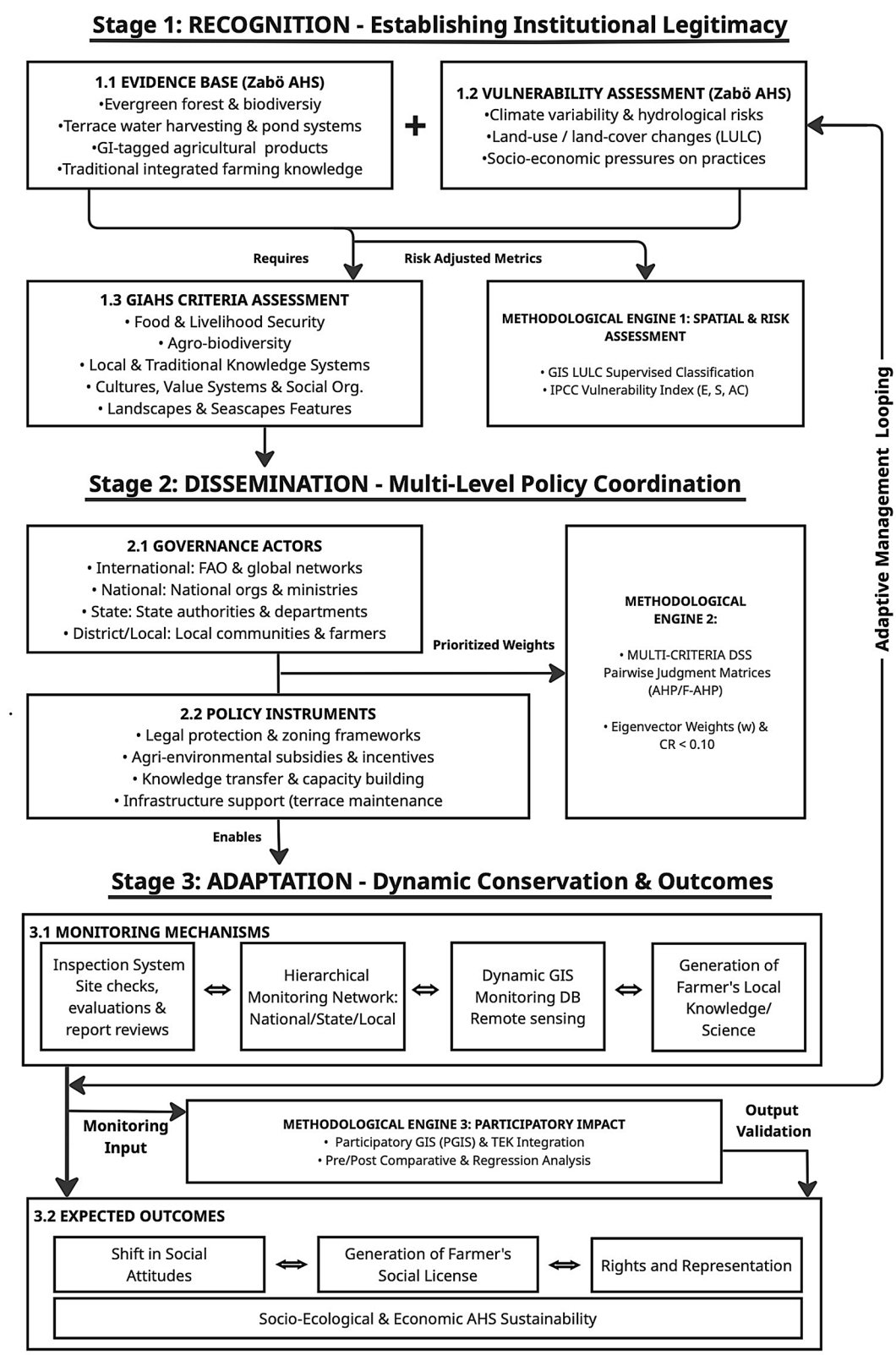
National Biodiversity Action Plan (NBAP). These policies at different levels need to be well-connected and work according to the multi-level governance for the multi-institutional and multi-disciplinary network. These policy instruments can be grouped into three functional clusters, regulatory instruments for formal recognition and protection, economic instruments for funding, market access, and compensation and capacity building instruments for knowledge transfer and community engagement.

At present, none of the reviewed schemes provide specific legal protection for Zabö terraces or catchments making GIAHS recognition particularly important as a potential primary regulatory mechanism. Extending capacity building instruments to support using engagement and knowledge transmission could also address the risk of outmigration and generational discontinuity. Figure 7 operationalizes this framework through three sequential stages and seven sequential sub-steps: evidence base, vulnerability assessment, GIAHS criteria assessment, governance actors, policy instruments, monitoring mechanisms, and expected outcomes.

The first three stages establish the evidence base and urgency for recognition, converging on institutional legitimacy. This legitimacy then engages governance actors, which feed into an AHP-based decision process that prioritizes regulatory, economic, and capacity-building instruments. The monitoring stage subsequently feeds updated environmental and socio-economic information back into the vulnerability assessment, creating a feedback mechanism through which policy priorities can be adjusted under changing conditions.

The monitoring flow chart for the policy recommendation involves a systematic approach. It begins with an inspection system that includes report examination, comprehensive evaluation, and site inspection. This monitoring network operates hierarchically through national organizations, state authorities, and local authorities. A dynamic monitoring system is in place, supported by a database and management system that facilitates the collection and analysis of data. Annual reports and surveys are conducted to track progress and identify areas needing improvement. Conservation measures are implemented based on the insights gained from this monitoring process, ensuring effective oversight and management of policy implementation at all levels of governance. Updated LULC and groundwater information generated through monitoring can re-enter the vulnerability assessment, allowing emerging pressures such as further catchment encroachment to inform subsequent policy priorities. This feedback mechanism represents the operational basis of dynamic conservation within the proposed framework. This structured flow chart ensures that policies are monitored and adjusted as needed to achieve sustainable development and environmental conservation objectives. To operationalize this multi-level governance structure, formal administrative tiers must be complemented by non-governmental and private sector actors. These administrative tiers are complemented by non-governmental and private-sector actors including the Chakhesang Women Multipurpose Cooperative Society, which already organizes women farmers across Phek district; the North Eastern Regional Agricultural Marketing Corporation (NERAMAC), which has authorized GI certification for other Naga crops (Naga Mircha, Naga Tree Tomato, Naga Sweet Cucumber) under the Ministry of Development of Northeast Region; the Krishi Vigyan Kendra (KVK), Phek, as the district's agricultural extension and research body; and Nagaland's network of Farmer Producer Companies/Organizations, whose participation is essential for translating policy instruments into ground-level implementation.

Recent work on collaborative design and social resilience in the Indian Sundarbans Delta shows that participatory processes can help connect community knowledge with institutional decision-making in climate-risk contexts (Majumdar & Ruj, 2026). This awareness is relevant to Zabö because the long-term conservation of agricultural heritage systems depends not only on ecological protection but also on shared learning, community-led documentation, and locally negotiated adaptation. In the framework's anticipated outcomes is the generation of what conservation and resource-governance literature terms a 'social license', the ongoing, community-level legitimacy and consent that sustains external policy intervention in a locally governed system, distinct from formal legal recognition alone. Social changes it brings ensure equitable opportunities for smallholder farmers by creating value-added behaviour. Since it is the key to agricultural development through the shift in social perspective through Rights and representation towards social-economic sustainability. Resources that are unique to the region need to be protected


Stage 1: RECOGNITION - Establishing Institutional Legitimacy
1.1 EVIDENCE BASE (Zabö AHS)
•Evergreen forest & biodiversiy
•Terrace water harvesting & pond systems
•GI-tagged agricultural products
•Traditional integrated farming knowledge
+
1.2 VULNERABILITY ASSESSMENT (Zabö AHS)
•Climate variability & hydrological risks
•Land-use / land-cover changes (LULC)
•Socio-economic pressures on practices
Requires
Risk Adjusted Metrics
1.3 GIAHS CRITERIA ASSESSMENT
• Food & Livelihood Security
• Agro-biodiversity
• Local & Traditional Knowledge Systems
• Cultures, Value Systems & Social Org.
• Landscapes & Seascapes Features
METHODOLOGICAL ENGINE 1: SPATIAL & RISK ASSESSMENT
• GIS LULC Supervised Classification
• IPCC Vulnerability Index (E, S, AC)
Stage 2: DISSEMINATION - Multi-Level Policy Coordination
2.1 GOVERNANCE ACTORS
• International: FAO & global networks
• National: National orgs & ministries
• State: State authorities & departments
• District/Local: Local communities & farmers
Prioritized Weights
METHODOLOGICAL ENGINE 2:
• MULTI-CRITERIA DSS Pairwise Judgment Matrices (AHP/F-AHP)
• Eigenvector Weights (w) & CR < 0.10
2.2 POLICY INSTRUMENTS
• Legal protection & zoning frameworks
• Agri-environmental subsidies & incentives
• Knowledge transfer & capacity building
• Infrastructure support (terrace maintenance
Enables
Stage 3: ADAPTATION - Dynamic Conservation & Outcomes
3.1 MONITORING MECHANISMS
Inspection System Site checks, evaluations & report reviews
Hierarchical Monitoring Network: National/State/Local
Dynamic GIS Monitoring DB Remote sensing
Generation of Farmer's Local Knowledge/ Science
Adaptive Management Looping
Monitoring Input
METHODOLOGICAL ENGINE 3: PARTICIPATORY IMPACT
• Participatory GIS (PGIS) & TEK Integration
• Pre/Post Comparative & Regression Analysis
Output Validation
3.2 EXPECTED OUTCOMES
Shift in Social Attitudes
Generation of Farmer's Social License
Rights and Representation
Socio-Ecological & Economic AHS Sustainability

◂**Fig. 7** Long-term AHS sustainability through GIAHS recognition: a conceptual framework

under the umbrella of indigenous practices. Changing climatic and social perspectives are the existing problems for the region in the long run. Therefore, this framework aims to mention holistic parameters for the Sustainability of Zabö in the long run.

Discussion

In considering the direction for future research, this study highlights the need for a multidirectional approach towards sustainability and socio-economic growth through the ZIFS, which has not been adequately addressed in the Kikruma region. It discusses Zabö's unique approach due to its sustainability towards agriculture, wildlife preservation, and effective water management by using an integrated method. Agroforestry may effectively gather rainwater and foster biodiversity, along with the preservation of knowledge transmission throughout generations and community cooperation (Govil et al., 2023). Proposed framework in this study also resonates with participatory conservation scholarship from eastern India, where locally grounded nature-based engagement has been linked to improved biodiversity knowledge and environmental awareness (Ghosal et al., 2023). In the Zabö context, such participatory events must be adopted through the tribal institutes and verified through field-based community consultation. It gives a higher yield and needs fewer natural and economical resources, including efficient use of water, a conventional way of sustainable farming, low cost, and eco-friendliness. However, its yield can be affected by a range of factors such as soil type, rain, anthropogenic changes, soil erosion, and climate change. Therefore, this research identified vulnerabilities, as we have seen in Sect. "Sustainability of the system", this region has changed drastically in the changing climate scenario. Therefore, this research proposed a conceptual framework that aims to address and improve the sustainability and socio-economic growth of this region. Thus, this research aligns ZIFS with the selection criteria for GIAHS. Further explores sustainability in the long run, by answering questions like why it needs to be recognized as a GIAHS and how. This area lacks awareness and capacity building towards the marketing of GI-tagged products in creating value chain reactions. Along with the validation of genetic resources. Therefore, it needs policy dialogue and dissemination towards the arising threats and challenges while keeping the economic benefits in check.

Moreover, this practice is limited to the lower elevation regions due to the abundance of water, which is concerning in terms of exhaust of run-off water in a rain-fed area. Alongside focusing on land Shaping is a major part of the management in the mountain region. During the land shaping project by the management of Nagaland Environment Protection and Economic Development Project (NEPED), farmers propose to use the natural resources due to labour intensity for trapping the topsoil in the Phek district. However, found that Indigenous approaches are more beneficial (IDRC, 1999). Being a rainfed region with a high percentage of slope due to the hilly spread in the region, the ultimate use of the catchments in a holistic way is the most sustainable way. Therefore, this practice demands to be protected since this region depends on both Surface and groundwater. Indigenous knowledge of the agrarian society, on-farm conservation, needs to be taken into consideration in similar terrain or climatic conditions. Although Zabö differs from mining-affected water landscapes in southwestern West Bengal, hydrosocial research from eastern India demonstrates that water governance is inseparable from power, access, livelihood security, and community resilience (Ruj, 2025). This supports the need to examine Zabö not only as an agricultural technique, but also as a community-managed hydrosocial system.

Policy recommendations to provide food and livelihood security and reduce risk, many AHS sites have built and modified their resilience to deal with climate variability and changes, resulting in dynamic conservation techniques and processes that enable the conservation of biodiversity and crucial ecosystem services. The depth and breadth of collected expertise in resource management and usage represent a priceless global treasure that must be encouraged, preserved, and given room to grow at the same time. Therefore, multi-governance policy dissemination is proposed. This region lacks a specific policy or an integrated policy dissemination for the preservation of the unique agricultural practices. This region has not been included in any of the national and state

policies for conservation of regional practices. Therefore, a multi-stakeholder policy framework has been suggested under the AHP way of policy dissemination and dynamic monitoring.

These findings have direct implications for three groups of practitioners. For policymakers, the criteria alignment evidence (as per the Table 1 and Fig. 7) provide a baseline template for highlighting Zabö system for nomination under India's national GIAHS pathways and existing schemes (PKVY, MOVCDNER, TDF, Watershed Development Fund) toward catchment and terrace maintenance rather than treating the system as a residual concern. For local community of Kikruma and other Chakhesang villages, formal recognition offers a route to securing labour and financial support for pond desilting and channel repair without displacing the community-based water sharing institutions that currently governs the system and any support mechanism should be designed in consultation with these institutions rather than imposing over them. For environmental planners working in comparable hilly ecosystems across North Eastern Himalayan region, the Zabö case demonstrates how gravity-fed, community-governed catchment systems can be viable climate adaptation strategy in rainfed, high-slope terrain, and offers a transferable methodology that can be scaled using the same evidence-to-outcome framework, adapted to local topography, institutions, and climatic vulnerability.

Conclusion

This study sought to examine how Zabö integrated farming system can be assessed as a potential GIAHS through a spatially grounded, dynamic conservation framework through proposing a framework for its long-term dynamic conservation. The observation suggests that the system aligns closely with FAO's GIAHS selection criteria across food and livelihood security, agrobiodiversity, traditional knowledge, social organization, and landscape criteria, that it faces measurable vulnerabilities from groundwater decline, soil erosion, and built-up area expansion, and that its recognition, dissemination and adaptation can be organized through the seven stages within a conceptual framework. The proposed framework organised through an analytical hierarchy process across international, national, state and local governance levels, offers a replicable structure for closing the current policy gap that excludes Zabö from national conservation strategies.

Ultimately, this study advances the analytical geography of agricultural heritage by conceptualizing Zabö as a co-evolving biocultural and hydrosocial landscape. By linking Indigenous knowledge systems and socio-ecological resilience with a structured recognition framework, we demonstrate how marginal mountain systems can bridge the gap between local resource governance and national policy recognition. In conclusion, it is crucial to recognize and promote indigenous farming traditions in India to address the social, environmental, and economic issues that rural populations face. The Indian subcontinent is complex and diverse. Therefore, it needs to disseminate region-specific planning for the AHS unique to its region. India can benefit at the local level in several ways by recognizing, maintaining, and mainstreaming these practices. Each site has different indicators alongside the similarities concerning GIAHS. Similarities such as the preservation of traditional wisdom, conservation of biodiversity, cultural identity, and social welfare. However, support for policy and research, socioeconomic benefit, conservation at a large scale, local economies, and genetic biodiversity depend on the region's uniqueness and the approach by the local or state government alongside the PPP. Additionally, acknowledging and recognizing the AHS as GIAHS in India can act as a role model for sustainable agriculture, fostering climate change resilience, preserving biodiversity, and fostering inclusive economic growth. This acknowledgement should not be interpreted as a method of preserving these systems in the past, but rather to encourage their adaptability to future problems while preserving cultural and environmental diversity. Acknowledging and promoting indigenous agricultural practices in India is a crucial step towards constructing a more sustainable and financially stable future for agricultural communities.

Future research should prioritize these directions. Quantitative field based hydrological monitoring of Zabö catchments and ponds is needed to validate the groundwater and recharge estimates since, current data is drawn from district level CGWB data. Comparative studies across other Chakhesang and neighbouring Naga villages will clarify how representative the Zabö practice of Kikruma case is for the wider practice and support a more robust GIAHS

nomination. Moreover, the participatory, community-based documentation of water sharing institutions will strengthen both the evidentiary basis for recognition and community ownership of the process. If these areas would be tested, the scalability of the proposed framework can be practically extended.

Limitations

Although this research aims to act as a comprehensive approach. However, it has its limitations. Despite offering a holistic framework for agricultural heritage conservation. The spatial and institutional insight are derived primarily from the Zabö system in Nagaland, meaning its broader policy applicability remains contingent on pilot testing and contextual adaptation in geomorphologically similar mountain regions.

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Author contributions K.Y.: Conceptualization, Methodology, Software, Resources, Writing—original draft, review & editing, Visualization. V.C.: Conceptualization, Supervision, Writing—review & editing. All authors reviewed the manuscript.

Data availability The data that support the findings of this study are available upon request from author.

Declarations

Conflict of interest The authors declare no conflict of interest.

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